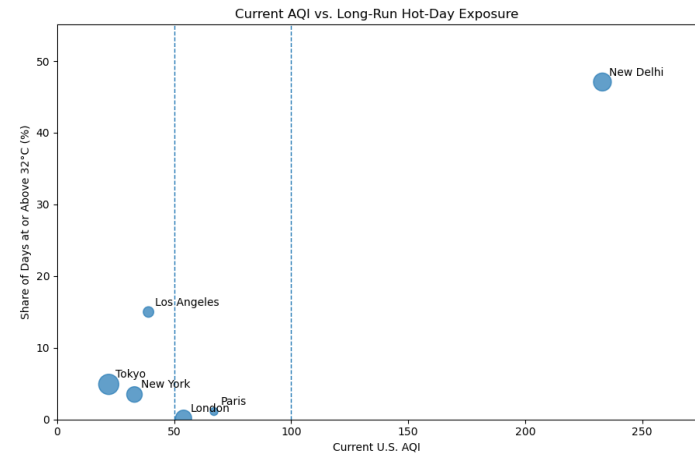


Global City Environmental Health Risk Analysis

Combining current air quality with long-run heat exposure for relocation decisions

Amy Tao & Victor Doddy · BUSN 32120 · Spring 2026



Current AQI vs. long-run heat exposure

6

selected cities

56,982

daily weather rows

12

SQL queries

2

modeling approaches

Executive answer: no single metric is enough

The practical question is not “Which city is best?” It is which cities combine different kinds of environmental concern: current air quality, long-run heat exposure, and recent heat trends.

City	Current AQI	Hot-day share	Trend	Read
New Delhi	233	47.1%	-3.0	Highest concern
Los Angeles	39	15.0%	+5.2	Heat concern
Tokyo	22	4.9%	+33.2	Trend watch
New York	33	3.5%	+11.0	Trend watch
Paris	67	1.1%	+1.4	AQI concern / low heat
London	54	0.2%	+0.8	Lower heat risk

Main finding

New Delhi is the only city in the sample that is extreme on both dimensions: current AQI and long-run heat exposure.

Second finding

Los Angeles looks relatively clean in the AQI snapshot, but it is still a meaningful heat-exposure case.

Trend watch

Tokyo and New York are not the hottest historically, but recent hot-day increases make them worth monitoring.

Data design: three sources, different grains

Dataset	Grain	Use in analysis
IQAir AirVisual	1 current row per city	AQI snapshot and current-risk dimension
Open-Meteo Archive	daily city-weather, 2000–2025	long-run heat, precipitation, wind, trend features
City context table	1 row per city	region, climate group, coastal status, population proxy

Check	Result
Missing values in AQI and weather	0 missing fields in shown checks
Duplicate city-date weather rows	0 rows
Invalid weather values	0 min>max, negative precipitation, negative wind
Date coverage	2000-01-01 to 2025-12-31 for all cities
Rows per city	9,497 daily rows × 6 cities = 56,982 rows

Interpretation caveat

The AQI data is a current snapshot, not a long-run air-quality average. The weather data provides the long-run history.

26 yrs

weather history

56,982

daily rows

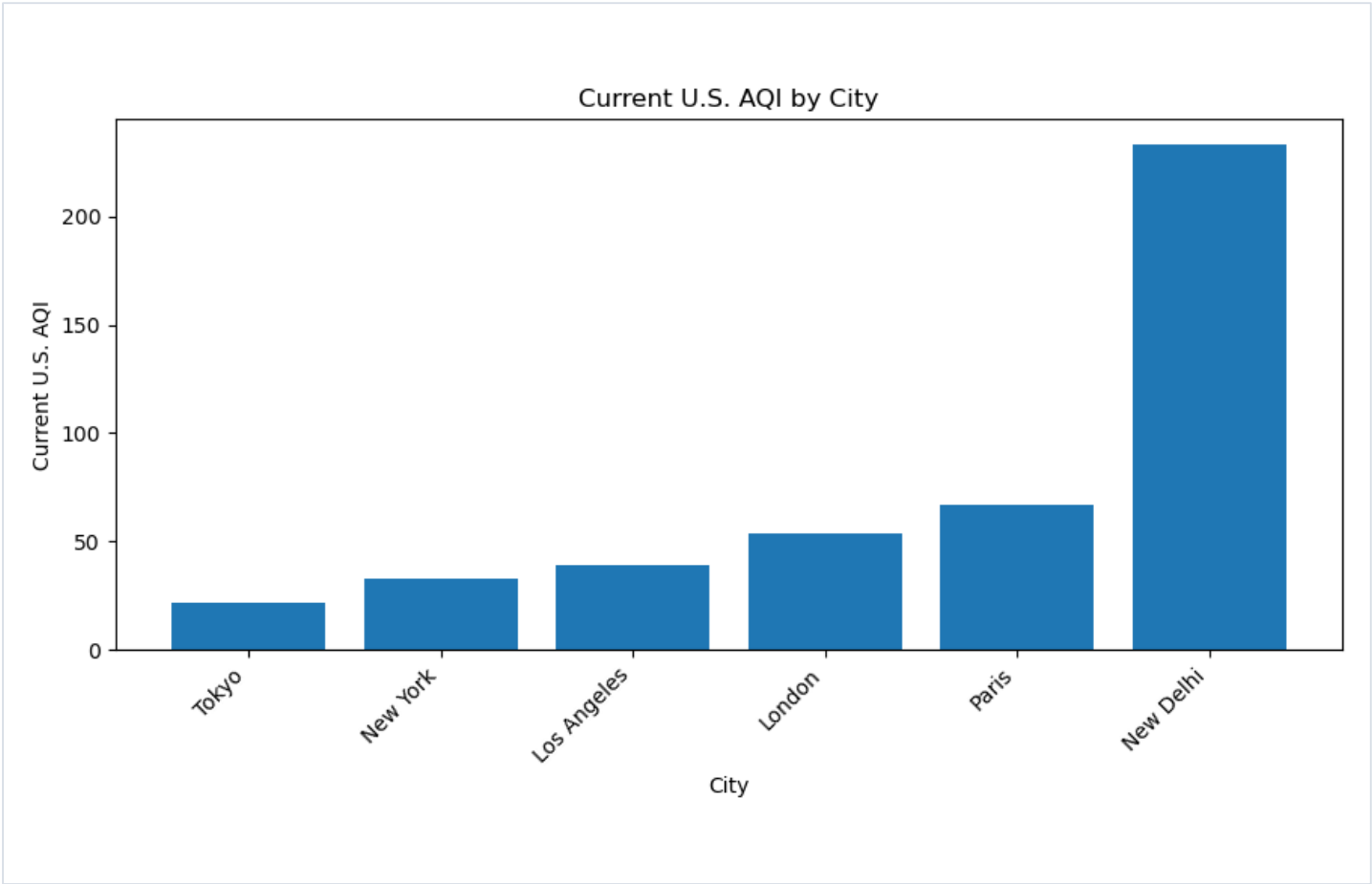
0

problem rows in validity checks

6

cities

Current AQI snapshot: New Delhi is a clear outlier

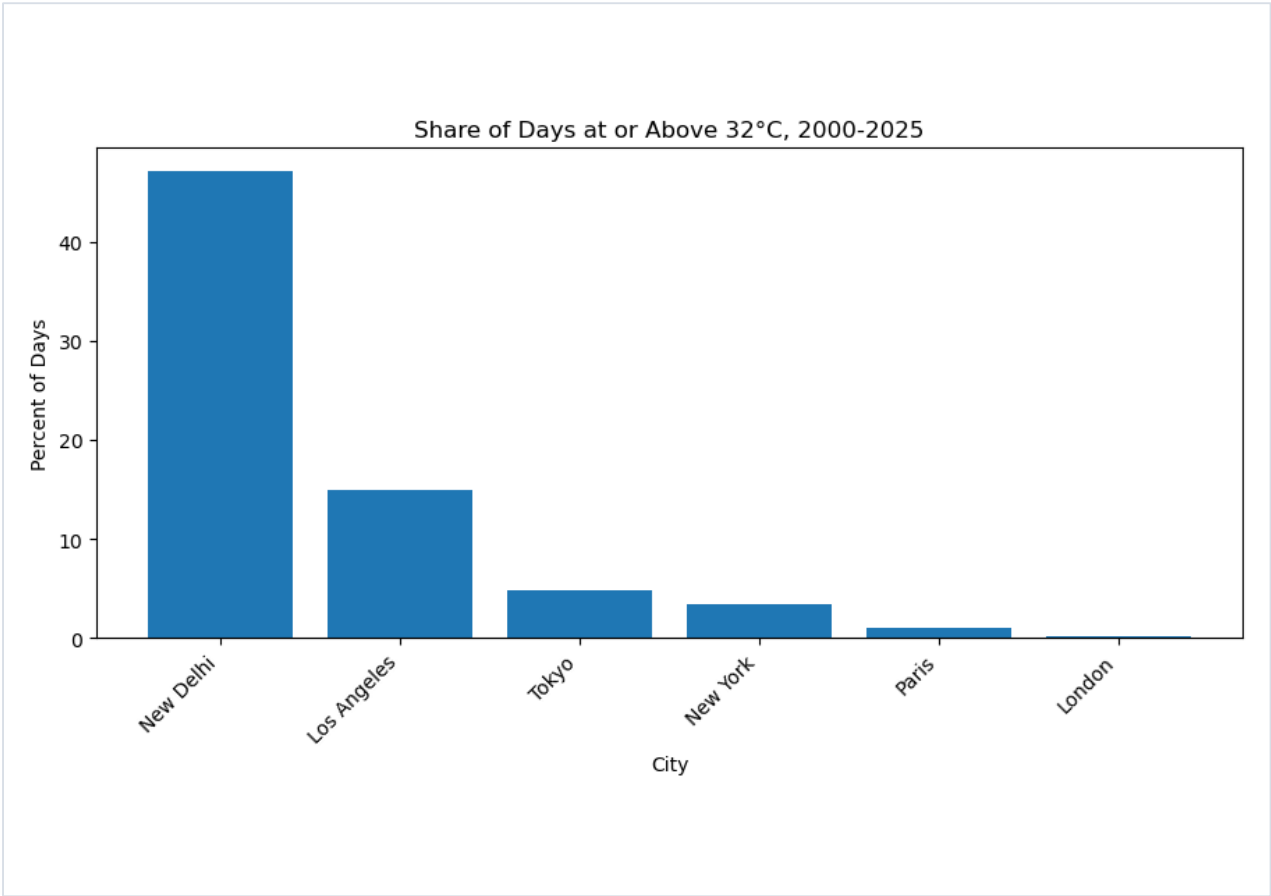


City	AQI	Category
New Delhi	233	Very unhealthy
Paris	67	Moderate
London	54	Moderate
Los Angeles	39	Good
New York	33	Good
Tokyo	22	Good

What this means

AQI alone would flag New Delhi most strongly. But AQI is only a current-state view; it needs to be paired with long-run heat exposure.

Long-run heat exposure: New Delhi and Los Angeles separate from the rest

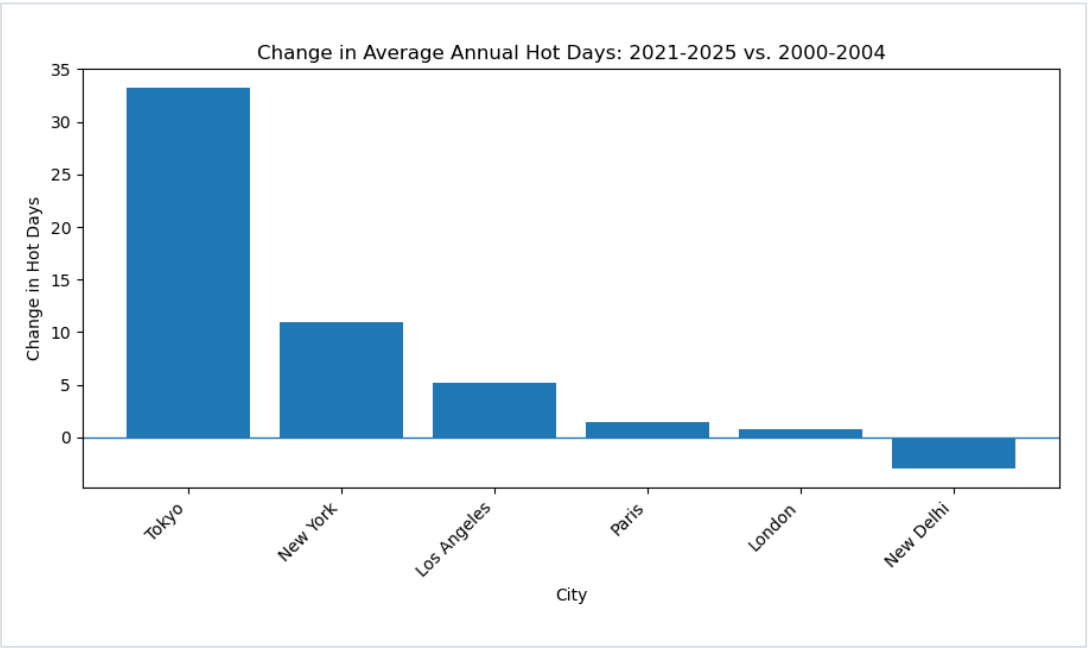
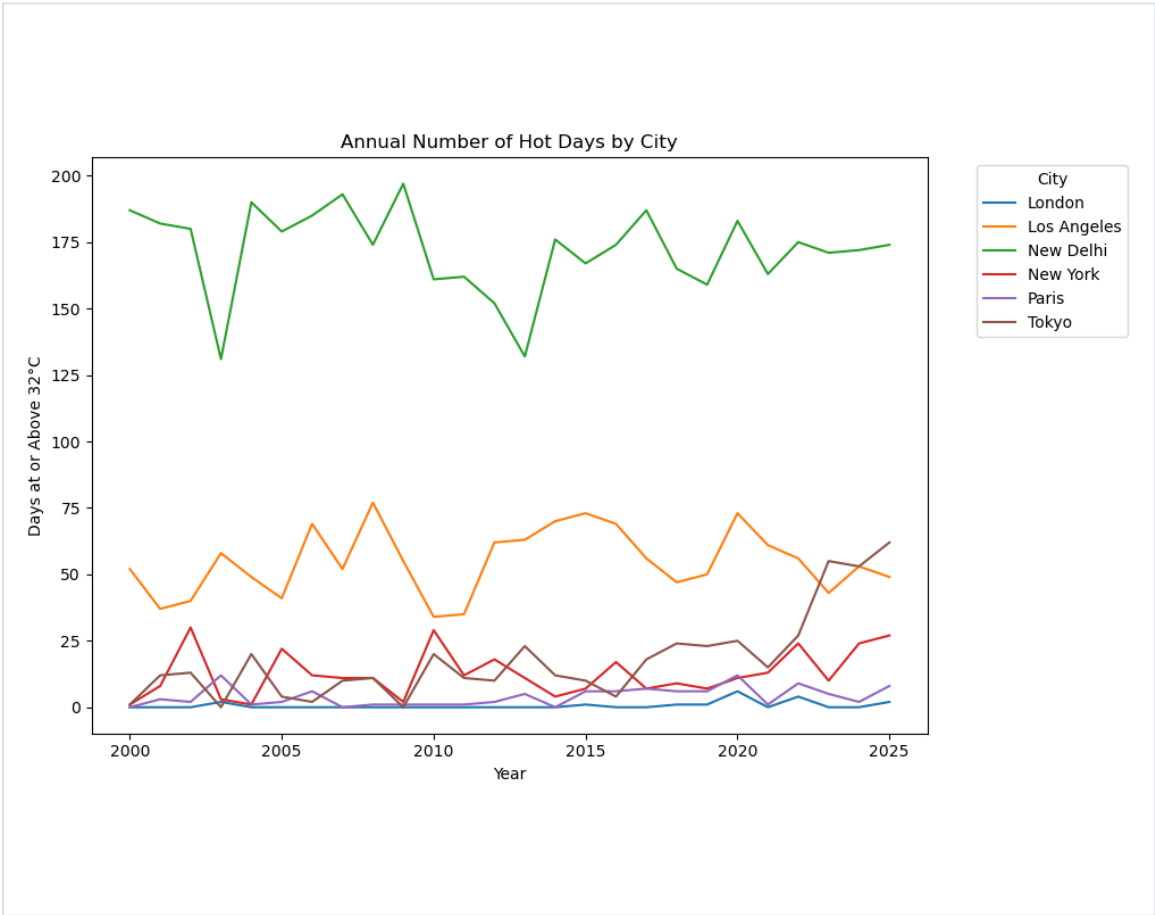


City	Hot-day share	Hottest day
New Delhi	47.1%	46.0°C
Los Angeles	15.0%	45.7°C
Tokyo	4.9%	39.3°C
New York	3.5%	40.4°C
Paris	1.1%	40.7°C
London	0.2%	37.9°C

Reading the chart

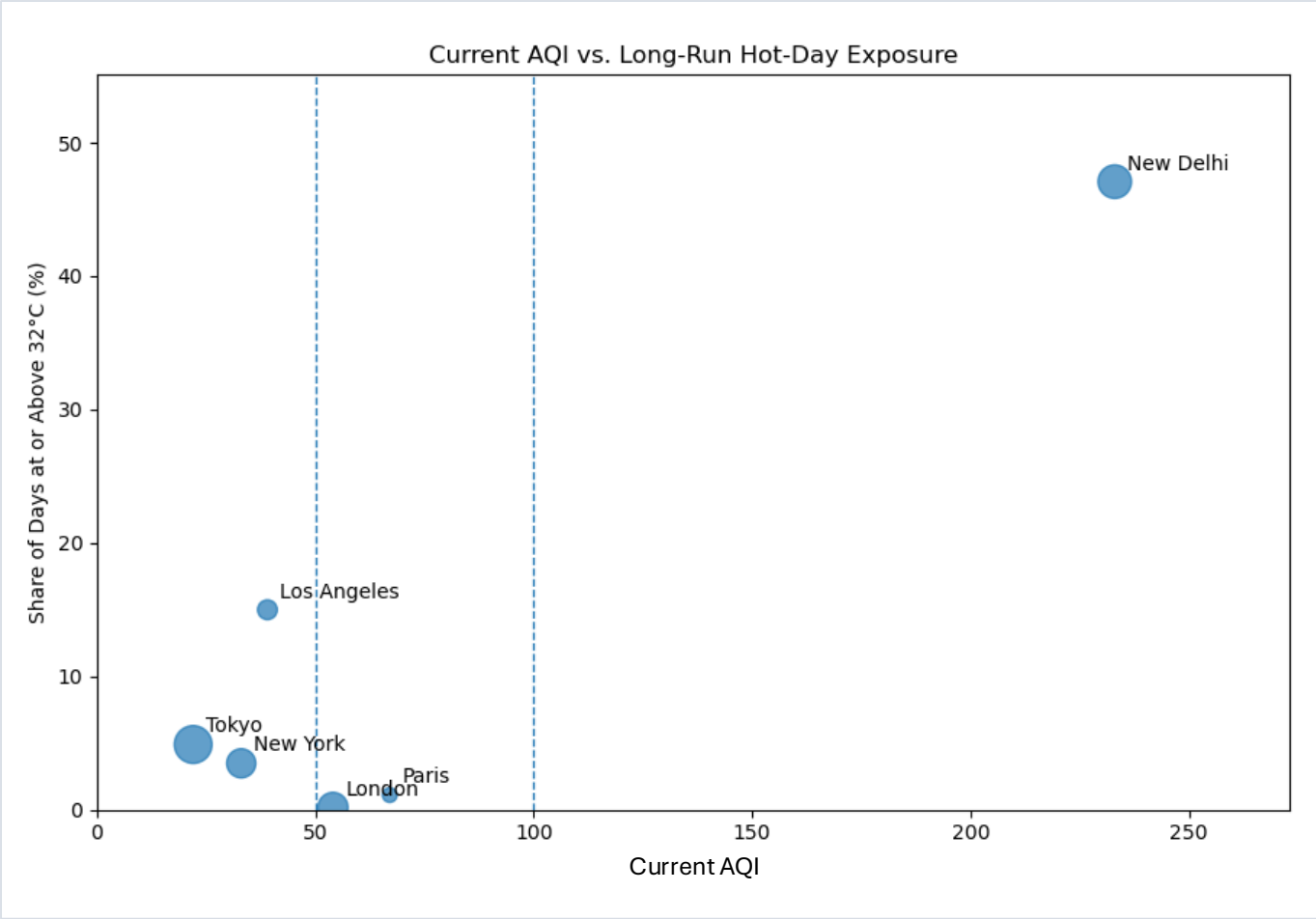
Hot days are defined as max temperature $\geq 32^{\circ}\text{C}$. New Delhi has a structurally different heat profile; Los Angeles is the next most exposed city.

Trend signal: Tokyo and New York are the watch cases



City	2000–04	2021–25	Change
Tokyo	9.2	42.4	+33.2
New York	8.6	19.6	+11.0
Los Angeles	47.2	52.4	+5.2
Paris	3.6	5.0	+1.4
London	0.4	1.2	+0.8
New Delhi	174.0	171.0	−3.0

Combined risk: AQI snapshot plus long-run heat exposure



City	AQI rank	Heatrank	Trend rank	Risk score
New Delhi	1.00	1.00	0.17	0.83
Los Angeles	0.50	0.83	0.67	0.67
Paris	0.83	0.33	0.50	0.57
Tokyo	0.17	0.67	1.00	0.53
New York	0.33	0.50	0.83	0.50
London	0.67	0.17	0.33	0.40

Risk score definition

Environmental risk score = 40% AQI rank + 40% heat-exposure rank + 20% recent trend rank. It is an interpretable index, not a medical claim.

SQL analysis: separate, auditable query file

The project includes a separate SQL script with 12 commented SQLite queries. The notebook writes cleaned Python data to SQLite, then verifies the query file runs end to end.

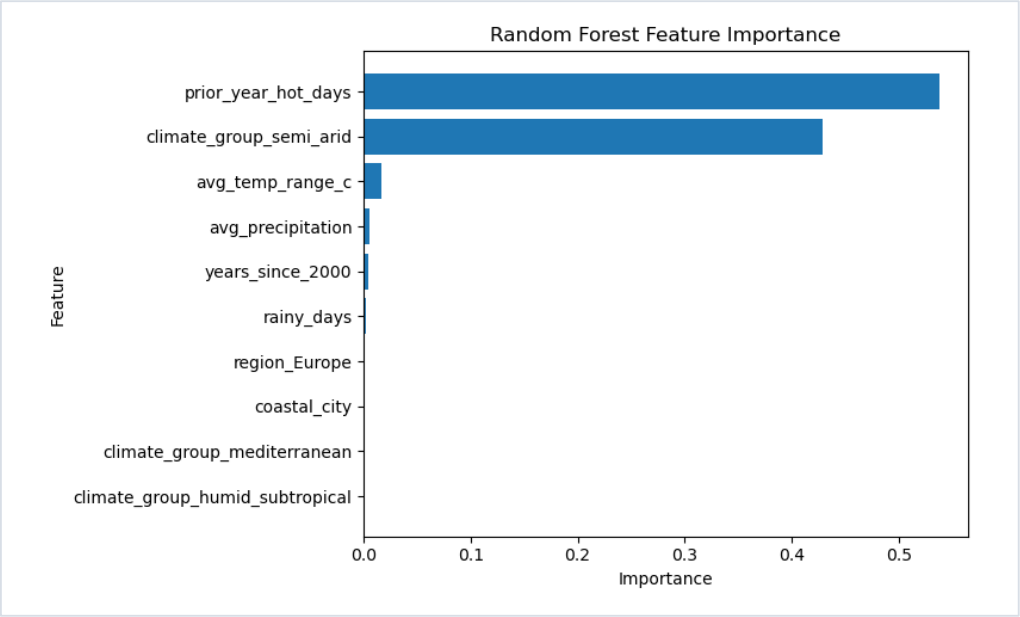
Query area	Requirement demonstrated	Finding used in analysis
AQI category counts	GROUP BY	3 Good, 2 Moderate, 1 Very Unhealthy
City ranking	WINDOW rank	New Delhi worst AQI; Tokyo best AQI snapshot
Above-average AQI	SUBQUERY	Only New Delhi exceeds sample AQI mean
Above-average heat	SUBQUERY	New Delhi and Los Angeles exceed heat-share average
Dual high risk	JOIN + subquery	New Delhi is sole city above both AQI and heat averages
Trailing trend	WINDOW average	Hot-day counts rising in Tokyo, New York, and Los Angeles

Why this matters

The SQL reproduces key EDA logic, joins the datasets, and creates an independent audit trail for the final findings.

Modeling annual hot days: strong in-sample checks, not a forecasting product

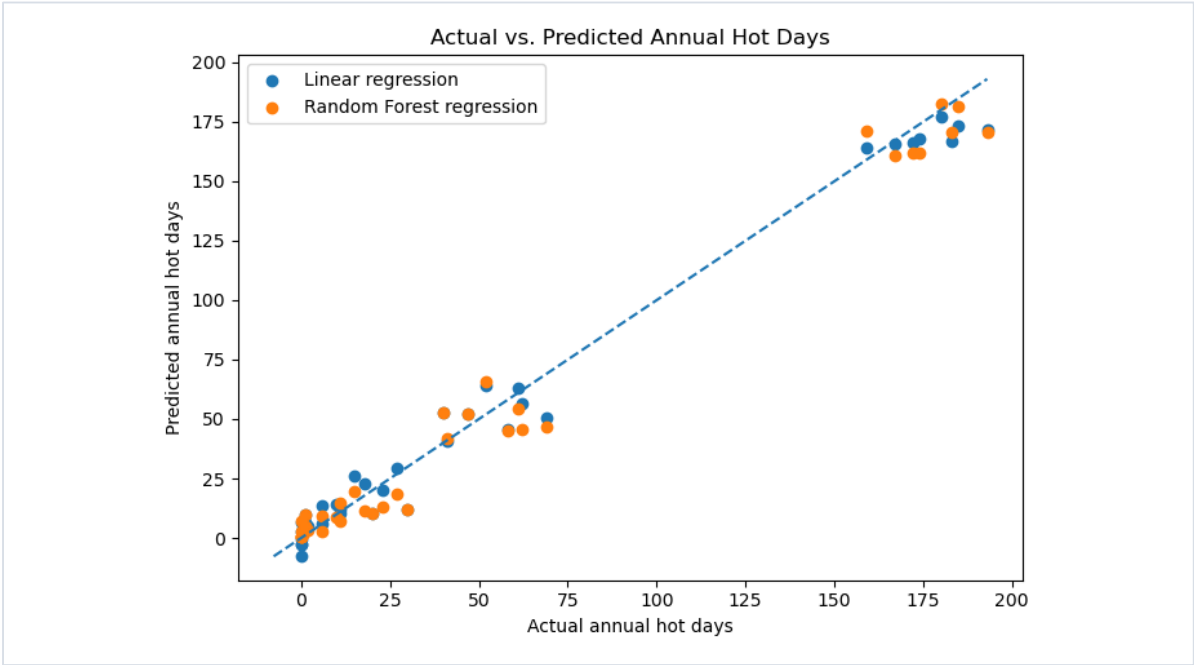
Model	MAE	RMSE	R ²	Interpretation
Linear regression	6.53	8.51	0.984	Simple baseline; slightly better metrics
Random Forest	7.27	9.37	0.980	Flexible ML check; useful feature importance



Interpretation

High R^2 reflects held-out city-years from the same six known cities. Feature importance concentrates in prior-year hot days and semi-arid climate group.

Key engineered features included hot-day flags, seasonal groupings, recent heat-trend measures, prior-year hot days, and the combined risk score.



Validation check: holding out entire cities changes the story

Held-out city	MAE	Actual avg hot days	Predicted avg
New Delhi	125.2	171.4	46.1
Los Angeles	50.2	54.9	105.1
Tokyo	36.6	18.6	55.1
New York	8.4	13.2	14.4
London	3.2	0.7	3.7
Paris	2.3	4.2	4.3

What the holdout shows

The model performs reasonably for cities similar to others in the sample. It fails on distinctive climates, especially New Delhi, because no training city has comparable heat exposure.

How this changes interpretation

The models are structured checks that the engineered features explain heat exposure within this sample. They are not production-ready tools for new global cities.

Presentation takeaway

High random-split R^2 shows usefulness within cities in the dataset, but the city-holdout table clearly demonstrates the limitations.

Recommendations for relocation-oriented decisions

Decision read	Cities	Why
Highest concern	New Delhi	Very unhealthy current AQI and highest long-run heat exposure
Heat concern	Los Angeles	Good AQI snapshot, but 15% hot-day share and high heat baseline
Trend watch	Tokyo, New York	Not hottest historically; recent hot-day counts increased materially
Lower heat exposure	London, Paris	Moderate AQI snapshot but low long-run hot-day exposure

Final answer

For this selected sample, New Delhi is the only clear dual-risk city. Los Angeles is the key “clean-air snapshot but high-heat” example. Tokyo and New York are future-monitoring cases.

Practical use: do not use one score by itself. Use the score to focus attention, then review the underlying AQI, long-run heat, and trend metrics city by city.

Limitations and future work

Limitations

- Current AQI is a point-in-time snapshot, not a long-run average.
- Six-city sample is intentionally selected and should not generalize globally.
- Random train/test split predicts held-out years from known cities, not new cities.
- Weather and AQI are environmental indicators, not direct health outcomes.

Future work

- Add historical air-quality data rather than one AQI snapshot.
- Add health outcomes and demographic vulnerability measures.
- Expand to more cities and more climate zones.
- Test generalization by holding out cities or regions.

Key takeaway

The main lesson is that a single environmental metric can be misleading. New Delhi is the clearest dual-risk case, but Los Angeles, Tokyo, and New York each show a different kind of environmental concern.

Questions

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